+39 3792239906
majidii.parsa@gmail.com

Parsa Majidi

Embedded Systems Engineer

Linkedin
github

Embedded Systems Engineer with expertise in Linux/Android BSP bring-up and low-level firmware across embedded Linux and Android platforms. At Kynetics, I work on board enablement, driver integration, and performance tuning, using oscilloscopes/logic analyzers and JTAG to diagnose hardware–software issues. Graduated M.Sc. in ICT (Smart Industry 4.0) with research interests in real-time embedded computing and efficient edge AI. Passionate about building reliable, production-ready systems from schematic to kernel to application.

Experience

Kynetics Inc. - Apprentice **Aug 2024 – Present**
Firmware | Embedded System Engineer | C/C++, Android/Linux BSPs, Firmware, Kernel, OTA Padova, IT

- Developed and integrated PD firmware in C for a custom iPad stand, increasing third-party USB-C hub interoperability from 40% to 90%.
- Ported and maintained Android (AOSP) and Linux BSPs, configuring bootloaders and kernel parameters for custom ARM-based hardware platforms
- Debugged device drivers for Ethernet, I2C, CAN, and Bluetooth; resolved low-level hardware-software conflicts using JTAG, oscilloscopes, and logic analyzers
- Ported and configured PipeWire, PulseAudio, and BlueALSA for custom Linux builds ; optimized Bluetooth A2DP/HFP profiles to achieve better audio perfomance.
- Upstreamed code to the spi-tools project to improve SPI bus utility for the broader developer community (Commit: 31c8a46)

Education

University of Padova, Italy | Supervisor: Prof. Andrea Triossi **Oct 2022 – October 2025**
M.S. ICT for Internet and the Internet - Smart Industry 4.0

- **Thesis:** Implementation of Tree Tensor Networks, a quantum-inspired machine learning algorithm, on the AI Engine of Versal adaptive SoCs for ultra-low latency applications
  - Vitis, Intrinsics APIs, AIE APIs, Quantum Machine Learning Algorithms, Versal Architecture, Multiply-Accumulate

University of Mazandaran, Iran **Oct 2018 – July 2022**
B.S. Electrical Engineering, GPA: 3.66/4.00

- **Thesis:** Driver Drowsiness Detection | Tensorflow lite, OpenCV, Image Processing, Raspberrypi, TinyML - Grade: 20/20

Projects

Porting and Developing Android 14 on Verdin iMX8MP

- Adapted and optimized Android 14 for the Verdin i.MX 8M Plus module
- Achieved 70–90% boot-time reduction by enabling and fine-tuning hibernation features
- Customized the Linux kernel, optimized bootloader settings, and integrated peripheral support

Neural Network Implementation on FPGA Aritx A7 **Link**

- Implementation of a Keras neural network using HLS4ML on an Artix-7 FPGA
- Integrated with a custom VHDL-based UART module for real-time serial communication

Programming the TurtleBot for Following a Path

- Team project for a line-following robot using STM32 microcontroller
- Ranked 1st out of 10 teams in Embedded Real-Time Control course by optimizing PID control for fastest line-tracking

Skills

Programming	C/C++, Python, Rust, Verilog, VHDL
Embedded Systems	Embedded Linux, Android/Linux BSP, FPGA, Yocto, QEMU, PCB
Debugging	JTAG, GDB, Logic Analyzers
Communication	Ethernet, I^2C, UART, SPI, CAN, Bluetooth, PulseAudio, BlueALSA, PipeWire
Dev Software	Vitis Unified IDE, Vivado, Altium Designer
Tools & Platforms	Docker, Git, Jira, Bitbucket, GitHub

Language

English	Professional Proficiency
Persian	Native
Italian	Basic